

Frontier Model Performance Across 20+ AI Benchmarks

A comparative analysis of 100+ models evaluated across Agentic, Frontier, Safety, and Multimodal capability domains

Agentic

Coding & Tool Use

Frontier

Reasoning & Knowledge

Safety

Alignment & Honesty

Multimodal

Audio & Video Understanding

100+

Models Evaluated

20+

Benchmarks

4

Domains

5+

Labs

20+ Benchmarks Spanning Four Capability Domains Evaluate 100+ Models

Scale SEAL Leaderboard Suite Overview

Agentic	Frontier	Safety	Multimodal
SWE Atlas – Test Writing SWE Atlas – Codebase QnA MCP Atlas (Tool Use) SWE-Bench Pro (Public) SWE-Bench Pro (Private)	Humanity's Last Exam (HLE) HLE – Text Only EnigmaEval MultiChallenge MultiNRC (Multilingual) SciPredict	MASK (Honesty) PropensityBench Fortress (Nat'l Security) Professional – Legal Professional – Finance Remote Labor Index	AudioMultiChallenge (Text Out) AudioMultiChallenge (Audio Out) VISTA Vision-Language VisualToolBench
5 benchmarks	6 benchmarks	6 benchmarks	4 benchmarks

GPT-5.4 Dominates Agentic Coding: Top Scores on SWE Atlas and SWE-Bench Pro

Top-3 models per benchmark | Scores with ±confidence intervals

Benchmark	#1 Model	#2 Model	#3 Model
SWE Atlas – Test Writing	#1 GPT-5.4-xHigh (Codex CLI) 44.36	#2 GPT-5.4-xHigh (Mini-SWE) 40.00	#3 GPT-5.3-Codex-Xhigh 38.98
SWE Atlas – Codebase QnA	#1 GPT-5.4-xHigh (Codex) 40.80	#2 GPT-5.4-xHigh (Mini-SWE) 36.30	#3 Claude Opus 4.6 (Claude Code) 33.30
MCP Atlas (Tool Use)	#1 Muse Spark 78.30	#2 Claude Opus 4.6 75.80	#3 Gemini 3.1 Pro Preview 73.90
SWE-Bench Pro (Public)	#1 GPT-5.4-pro (xHigh) 59.10	#2 Muse Spark 55.00	#3 Claude Opus 4.6 51.90
SWE-Bench Pro (Private)	#1 Claude Opus 4.6 47.10	#2 Muse Spark 44.70	#3 GPT-5.4-pro (xHigh) 43.40

Lab color coding:

OpenAI

Anthropic

Google

Muse Spark

GPT-5.4-Pro Leads Humanity's Last Exam at 44.3% — Scores Remain Below 50%

Top models on key frontier reasoning benchmarks | All scores well below saturation ceiling

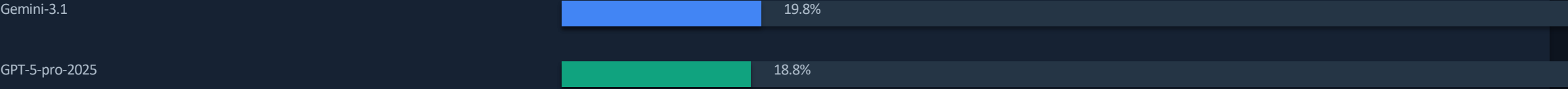
Humanity's Last Exam (HLE)

Top score 44.3% — ceiling at 100%



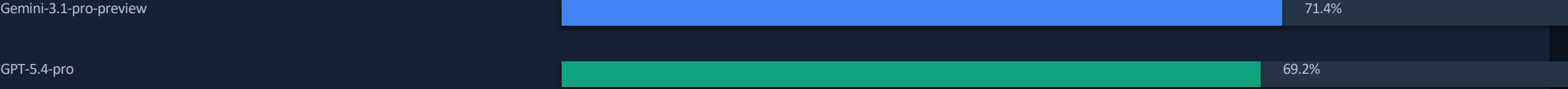
EnigmaEval

Top score 23.8% — remains highly unsaturated



MultiChallenge

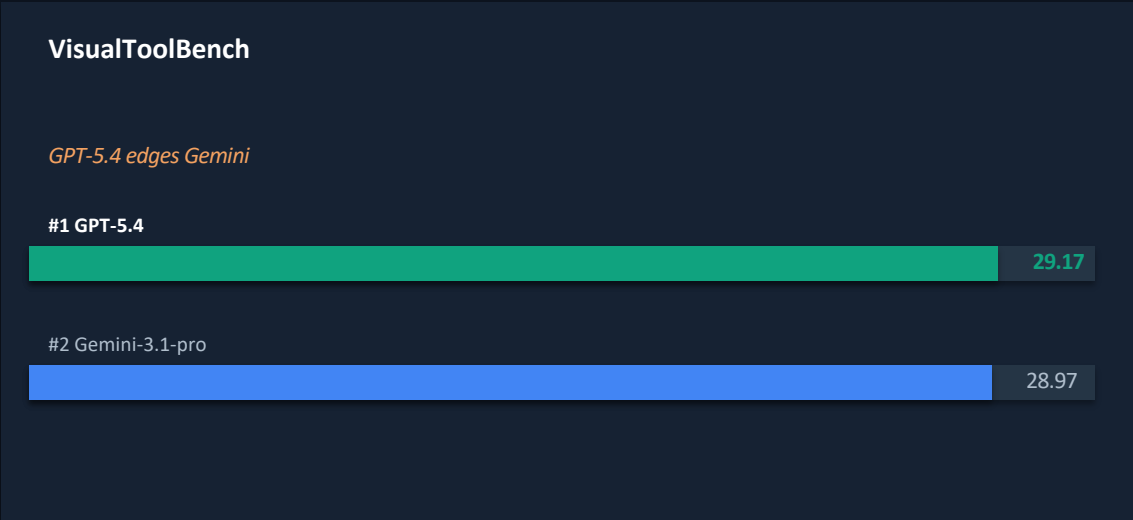
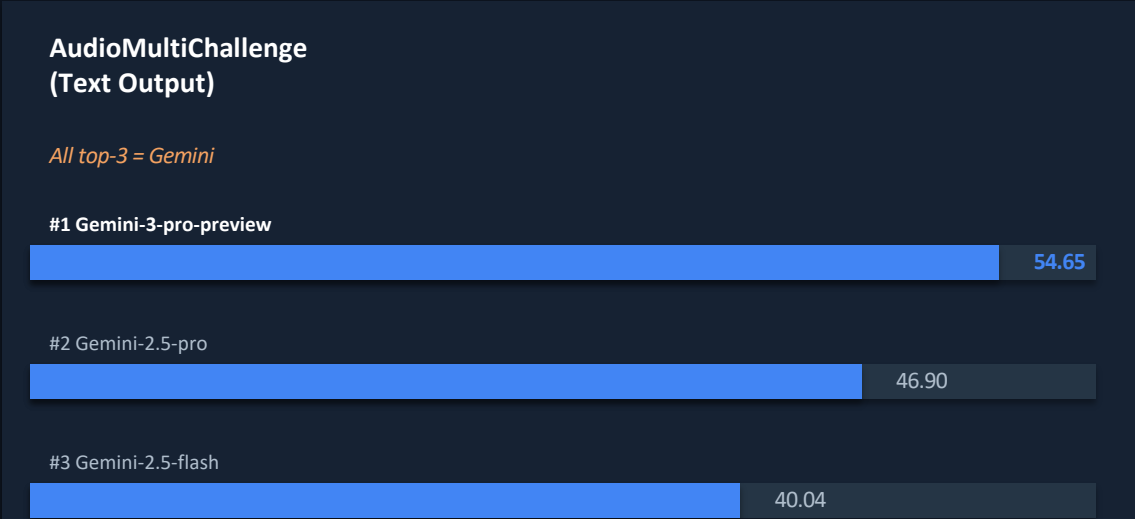
Muse Spark tops at 75.5%



Benchmark Saturation Signal: HLE top score = 44.3% | EnigmaEval top = 23.8% | SciPredict top = 25.3% — frontier models still have substantial room for improvement

Gemini Models Sweep Audio and Vision-Language Benchmarks

Google's Gemini variants hold top positions across all four multimodal leaderboards



Gemini Dominance: Holds all top-3 positions in AudioMultiChallenge (Text Output) | Leads VISTA | GPT-5.4 wins VisualToolBench by a narrow margin (29.17 vs 28.97)

Claude Models Score 95–96% on MASK Honesty; Fortress Scores Stay Low

Safety alignment varies dramatically by task type — near-ceiling honesty vs. critically low national security scores

MASK — Honesty Benchmark

Near-ceiling performance | Higher = better

Claude Opus 4.6



Claude Sonnet 4



Claude also leads Professional Reasoning:
#1 Legal: 52.27% · #1 Finance: 53.28%

Fortress & PropensityBench — Critical Low Scores

Fortress = National Security tasks | Lower PropensityBench = better refusal behavior

Best Fortress Score



Claude Sonnet 4 (Fortress)



o3-2025-04-16 (PropensityBench; lower=better)



Key Contrast: MASK honesty = 96% (near ceiling)
vs. Fortress national security = 8–13 (critically low)
— Safety alignment is highly task-dependent

OpenAI, Google, and Anthropic Each Claim #1 Across Different Domains

Heatmap: which lab's model ranks #1 on each benchmark | Muse Spark highlighted as new entrant

Benchmark	Domain	#1 Lab	#1 Model	Score
SWE Atlas – Test Writing	Agentic	OpenAI	GPT-5.4-xHigh (Codex CLI)	44.4
SWE Atlas – Codebase QnA	Agentic	OpenAI	GPT-5.4-xHigh (Codex)	40.8
MCP Atlas (Tool Use)	Agentic	Muse Spark	Muse Spark	78.3
SWE-Bench Pro (Public)	Agentic	OpenAI	GPT-5.4-pro (xHigh)	59.1
SWE-Bench Pro (Private)	Agentic	Anthropic	Claude Opus 4.6	47.1
Humanity's Last Exam	Frontier	OpenAI	GPT-5.4-pro	44.3
HLE (Text Only)	Frontier	OpenAI	GPT-5.4-pro	45.3
EnigmaEval	Frontier	OpenAI	GPT-5.4-pro	23.8
MultiChallenge	Frontier	Muse Spark	Muse Spark	75.5
MultiNRC (Multilingual)	Frontier	OpenAI	GPT-5-pro	65.2
SciPredict	Frontier	Google	Gemini-3-pro-preview	25.3
MASK (Honesty)	Safety	Anthropic	Claude Opus 4.6	96.3
Fortress (Nat'l Security)	Safety	N/A	Best score (low)	8.2
Prof. Reasoning – Legal	Safety	Anthropic	Claude Opus 4.6	52.3
Prof. Reasoning – Finance	Safety	Anthropic	Claude Opus 4.6	53.3
AudioMultiChallenge (Text)	Multimodal	Google	Gemini-3-pro-preview	54.7
AudioMultiChallenge (Audio)	Multimodal	Google	Gemini-3.1-flash-live	36.1
VISTA (Vision-Language)	Multimodal	Google	Gemini 2.5 Pro Exp	54.7
VisualToolBench	Multimodal	OpenAI	GPT-5.4	29.2

Lab colors: OpenAI Google Anthropic Muse Spark ★ New

Key Takeaways

No Single Model Wins Everything — Specialization Is the New Frontier

GPT-5.4 Leads Agentic Coding & Frontier Reasoning

Dominates SWE Atlas (44.4%), SWE-Bench Pro public (59.1%), HLE (44.3%), and EnigmaEval (23.8%). Clear #1 across software engineering and multi-step reasoning tasks.

Gemini Dominates Audio & Vision Benchmarks

Sweeps all top-3 spots on AudioMultiChallenge Text Output (54.7%→40%), leads Audio Output (36.1%), and tops VISTA vision-language (54.7%). GPT-5.4 wins VisualToolBench by 0.2 points.

Claude Leads Safety, Honesty & Professional Reasoning

Near-ceiling MASK honesty scores (96.3% / 96.1%) plus #1 in Legal (52.3%) and Finance (53.3%) professional reasoning. All models score critically low on Fortress (8–13).

Key Benchmarks Remain Far from Saturation

Humanity's Last Exam top = 44.3% | EnigmaEval top = 23.8% | SciPredict top = 25.3%. Frontier models still have substantial room for improvement across complex reasoning tasks.

Muse Spark Emerges as a Strong New Entrant

Top-3 on 6+ benchmarks: #1 MultiChallenge (75.5%), #1 MCP Atlas Tool Use (78.3%), #2 SWE-Bench Pro Public (55.0%), #2 HLE (40.6%). A meaningful new competitor across coding, tool use, and reasoning.